

**311. Step I:**  $6 \times 3 + 6^2 = 18 + 36 = 54.$

**Step II:**  $(54 \div 5) + 6 = 10.8 + 6 = 16.8.$

**Step III:**  $(16.8 \div 3)^2 = (5.6)^2 = 31.36.$

**312. Step I:**  $2.5 \times 3 + (2.5)^2 = 7.5 + 6.25 = 13.75.$

**Step II:**  $(13.75 \div 5) + 6 = 2.75 + 6 = 8.75.$

**Step III:**  $(8.75 \div 3)^2 = (2.917)^2 = 8.508 \approx 8.51.$

**313. Step I:**  $5 \times 3 + 5^2 = 15 + 25 = 40.$

**Step II:**  $(40 \div 5) + 6 = 8 + 6 = 14.$

**Step III:**  $(14 \div 3)^2 = (4.67)^2 \approx 21.80.$

**314.**  $2 \wedge a = a \wedge 3 \Rightarrow 2 \times 2 + a = 2 \times a + 3$   
 $\Rightarrow 4 + a = 2a + 3 \Rightarrow a = 1.$

**315.** 
$$\left[ \frac{(5 \oplus 7) + (4 \oplus 4)}{3(5 \oplus 5) - (15 \oplus 11) - 3} \right] = \left[ \frac{(5 + 7) + 4^2}{3 \times 5^2 - 2 \times 15 - 3} \right]$$

$$= \left[ \frac{12 + 16}{75 - 30 - 3} \right] = \frac{28}{42} = \frac{2}{3}.$$

**316.** Thickness of snow at 3 p.m. =  $\left[ 2\frac{1}{4} + \left( 3 \times 1\frac{1}{4} \right) \right]$  inches  
 $= \left( \frac{9}{4} + 3 \times \frac{5}{4} \right)$  inches  
 $= \left( \frac{9}{4} + \frac{15}{4} \right)$  inches  
 $= \left( \frac{24}{4} \right)$  inches = 6 inches.

**317.** In 2 min, the squirrel climbs  $(6 - 4)$  m = 2 m.

In  $(2 \times 57)$ , i.e., 114 min, the squirrel climbs  $(2 \times 57)$  m = 114 m.

In 115 min, the squirrel climbs  $(114 + 6)$  m = 120 m to reach the top.

**318.** For the money to grow 3 times, Ram needs to deposit ₹  $(3000 - 1000) = ₹ 2000.$

A net amount of ₹  $(750 - 500) = ₹ 250$  is deposited in 1 month.

So, a net amount of ₹ 2000 will be deposited in  $\left( \frac{2000}{250} \right) = 8$  months.

**319.** In 11 days, the height of the plant increases by  $(11 - 6)$  cm i.e., 5 cm.

In  $(11 \times 18)$  i.e., 198 days, the height of the plant increases by  $(5 \times 18)$  i.e., 90 cm.

In the next 10 days, the height of the plant increases by another 10 cm.

So, in 208 days, the height of the plant will increase by  $(90 + 10) = 100$  cm or 1 m and become 2 m.

**320.** Number of steps =  $\left( 10 \div \frac{1}{2} \right) = 20.$

To reach the top, the ant covers the height of 20 steps and the width of 19 steps because the moment it ascends the height of 20th step, it reaches the top.

$\therefore$  Distance travelled by the ant

$$= \left[ \left( \frac{1}{2} \times 20 \right) + (1 \times 19) \right] \text{ ft} = (10 + 19) \text{ ft} = 29 \text{ ft}.$$

**321.** Required sum =  $\frac{1}{2} + \left( \frac{1}{3} + \frac{2}{3} \right) + \left( \frac{1}{4} + \frac{2}{4} + \frac{3}{4} \right) + \left( \frac{1}{5} + \frac{2}{5} + \frac{3}{5} + \frac{4}{5} \right) + \dots$   
 $+ \left( \frac{1}{100} + \frac{2}{100} + \frac{3}{100} + \dots + \frac{99}{100} \right)$

$$= \frac{1}{2} + \frac{2}{2} + \frac{3}{2} + \frac{4}{2} + \dots + \frac{99}{2}$$

$$= \frac{1}{2} \times (1 + 2 + 3 + \dots + 99)$$

$$= \frac{1}{2} \times \frac{99 \times 100}{2} = 2475.$$

**322.** Given exp. =  $\frac{(1-x^2)(1+x^2)}{(1+x)} \times \frac{x}{1+x^2} \times \frac{1}{x(1-x)}$   
 $= \frac{(1-x)(1+x)(1+x^2)x}{(1+x)(1+x^2)x(1-x)} = 1.$

**323.** Given exp. =  $\left( \frac{17}{15} \right)^2 + \left( \frac{2}{15} \right)^2 - 2 \times \frac{17}{15} \times \frac{2}{15} = \left( \frac{17}{15} - \frac{2}{15} \right)^2 = 1.$

**324.** Given exp. =  $\left( 13\frac{3}{4} \right)^2 + \left( 5\frac{1}{4} \right)^2 + 2 \times 13\frac{3}{4} \times 5\frac{1}{4}$   
 $= \left( 13\frac{3}{4} + 5\frac{1}{4} \right)^2 = \left( \frac{55}{4} + \frac{21}{4} \right)^2 = \left( \frac{76}{4} \right)^2 = (19)^2 = 361.$

**325.** Given exp. =  $\frac{(75983)^2 - (45983)^2}{75983 - 45983}$   
 $= \frac{(75983 - 45983)(75983 + 45983)}{(75983 - 45983)}$   
 $= 75983 + 45983 = 121966.$

**326.** Given exp. =  $\frac{a^2 - b^2}{a - b}$ , (where  $a = 13\frac{4}{7}$ ,  $b = 8\frac{3}{7}$ )  
 $= a + b = 13\frac{4}{7} + 8\frac{3}{7} = \frac{95}{7} + \frac{59}{7} = \frac{154}{7} = 22.$

**327.**  $\frac{(a-b)^2 - (a+b)^2}{-4a} = \frac{x}{y} \Leftrightarrow \frac{(a^2 + b^2 - 2ab) - (a^2 + b^2 + 2ab)}{-4a}$   
 $= \frac{x}{y}$   
 $\Leftrightarrow \frac{-4ab}{-4a} = \frac{x}{y} \Leftrightarrow b = \frac{x}{y} \Leftrightarrow x = yb.$

**328.** Given exp. =  $\frac{\left( \frac{7}{8} + \frac{5}{6} \right)^2}{\left( \frac{7}{8} \right)^2 - \left( \frac{5}{6} \right)^2} = \frac{\left( \frac{7}{8} + \frac{5}{6} \right)^2}{\left( \frac{7}{8} + \frac{5}{6} \right) \left( \frac{7}{8} - \frac{5}{6} \right)}$   
 $= \frac{\left( \frac{7}{8} + \frac{5}{6} \right)}{\left( \frac{7}{8} - \frac{5}{6} \right)} = \left( \frac{41}{24} \times 24 \right) = 41.$

**329.** Given exp. =  $(1998 + 1997)(1998 - 1997) + (1996 + 1995)(1996 - 1995) + (1994 + 1993)(1994 - 1993)$   
 $= 1998 + 1997 + 1996 + 1995 + 1994 + 1993 = 11773.$